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MODEL QUESTION PAPER

COMPLEX ANALYSIS

DECEMBER 2016

1(a) Show that $u = y^3 - 3x^2y$ is harmonic & hence find its harmonic conjugate

(b) Find the image of $|z - \frac{1}{2}| \leq \frac{1}{2}$ under the transformation $w = \frac{1}{z}$

Also find the fixed points of the transformation $w = \frac{1}{z}$

Ans: (a)

$$u = y^3 - 3x^2y$$

$$u_x = -6xy$$

$$u_{yy} = 6y$$

$$u_y = 3y^2 - 3x^2$$

$$u_{xy} = -6x$$

$$u_{xx} = -6y$$

$$u_{yx} = -6x$$

They are continuous everywhere

$$\text{Also, } u_{xx} + u_{yy} = -6y + 6y = 0$$

$\therefore u = y^3 - 3x^2y$ is a harmonic function

Let v be the harmonic conjugate of u .

Then it satisfies C-R equation, $u_x = v_y$ & $u_y = -v_x$

$$u_x = v_y \Rightarrow v_y = -6xy$$

$$\Rightarrow v = \int -6xy \, dy = -\frac{6xy^2}{2} + g(x)$$

$$\Rightarrow v = -3xy^2 + g(x)$$

$$u_y = -v_x \Rightarrow 3y^2 - 3x^2 = -[-3y^2 + g'(x)]$$

$$3y^2 - 3x^2 = 3y^2 - g'(x)$$

$$g'(x) = 3x^2$$

$$g(x) = x^3 + c$$

$$\therefore v = -3xy^2 + x^3 + c$$

(b) Answer in NB, Module - 3, pg no: 68

To find fixed points

$$w = z \Rightarrow z = \frac{1}{z} \Rightarrow z^2 = 1 \Rightarrow z = \pm 1$$

$\therefore$ fixed points are $+1$ & -1

2. Define an analytic function

Ans: A function $f(z)$ is said to be analytic in a domain D if $f(z)$ is defined & differentiable at all points of D . The function $f(z)$ is said to be analytic at a point $z = z_0$ in D if $f(z)$ is analytic in a neighbourhood of z_0 .

3. (a) Show that $f(z) = e^{-x} \cos y - i e^{-x} \sin y$ is differentiable everywhere. Find its derivative

(b) Find the image of the lines $x = c$ and $y = k$, where c & k are constants, under the transformation $w = \sin z$

Ans: (a) $f(z) = e^{-x} \cos y - i e^{-x} \sin y = e^{-x} [\cos y - i \sin y] = e^{-x} \cdot e^{-iy}$

$$f(z) = e^{-x-iy} = e^{-(x+iy)} = e^{-z}$$

$$f(z) = e^{-z}$$

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z+\Delta z) - f(z)}{\Delta z}$$

$$= \lim_{\Delta z \rightarrow 0} \frac{e^{-(z+\Delta z)} - e^{-z}}{\Delta z}$$

$$= \lim_{\Delta z \rightarrow 0} \frac{e^{-z} \cdot e^{-\Delta z} - e^{-z}}{\Delta z}$$

$$= \lim_{\Delta z \rightarrow 0} \frac{e^{-z} - e^{-z} e^{\Delta z}}{\Delta z e^{\Delta z}}$$

$$= \lim_{\Delta z \rightarrow 0} \frac{(e^{\Delta z} - 1) \cdot \frac{-e^{-z}}{e^{\Delta z}}}{\Delta z}$$

$$f'(z) = 1 \times \frac{-e^{-z}}{e^0} = -e^{-z}$$

$$e^{i\theta} = \cos \theta + i \sin \theta$$

$$e^{-i\theta} = \cos \theta - i \sin \theta$$

$$z = x + iy$$

$$\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1$$

(b) $w = \sin z$

put $z = x + iy$, $w = u + iv$

$$u + iv = \sin(x + iy)$$

$$u + iv = \sin x \cosh y + i \cos x \sinh y$$

$$u + iv = \sin x \cosh y + i \cos x \sinh y$$

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$\cos iy = \cosh y$$

$$\sin iy = i \sinh y$$

Comparing, $u = \sin x \cosh y$, $v = \cos x \sinh y$ — (1)

$$\cosh^2 y - \sinh^2 y = 1$$

From (1), $\cosh y = \frac{u}{\sin x} \Rightarrow \cosh^2 y = \frac{u^2}{\sin^2 x}$

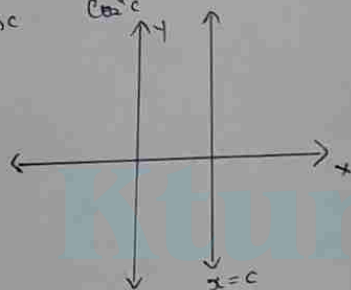
$\sinh y = \frac{v}{\cos x} \Rightarrow \sinh^2 y = \frac{v^2}{\cos^2 x}$

$\therefore \cosh^2 y - \sinh^2 y = 1 \Rightarrow \frac{u^2}{\sin^2 x} - \frac{v^2}{\cos^2 x} = 1$

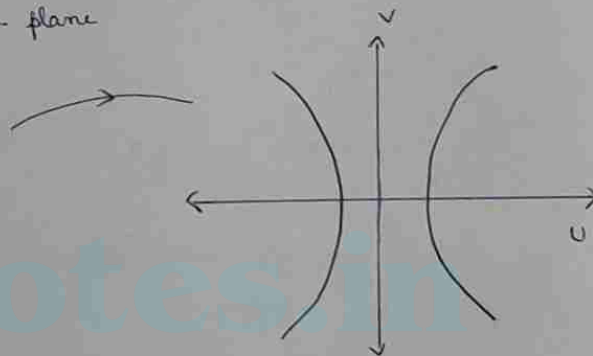
$x = c \Rightarrow \frac{u^2}{\sin^2 c} - \frac{v^2}{\cos^2 c} = 1$

ie, the image of the line $x=c$ in the z -plane is the hyperbola

$\frac{u^2}{\sin^2 c} - \frac{v^2}{\cos^2 c} = 1$ in the w -plane



z -plane



w -plane

$$\sin^2 x + \cos^2 x = 1$$

From (1), $\sin x = \frac{u}{\cosh y} \Rightarrow \sin^2 x = \frac{u^2}{\cosh^2 y}$

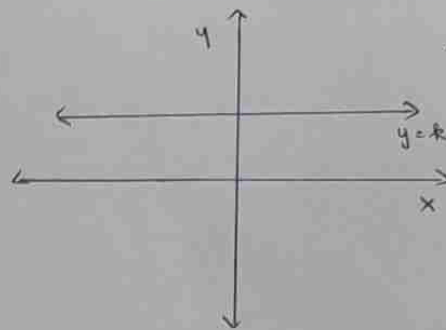
$\cos x = \frac{v}{\sinh y} \Rightarrow \cos^2 x = \frac{v^2}{\sinh^2 y}$

$\therefore \sin^2 x + \cos^2 x = 1 \Rightarrow \frac{u^2}{\cosh^2 y} + \frac{v^2}{\sinh^2 y} = 1$

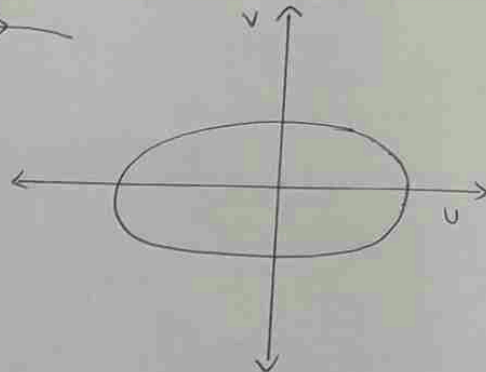
$y = k \Rightarrow \frac{u^2}{\cosh^2 k} + \frac{v^2}{\sinh^2 k} = 1$

ie, the image of the line $y=k$ in the z -plane is the parabola

$\frac{u^2}{\cosh^2 k} + \frac{v^2}{\sinh^2 k} = 1$ in the w -plane.



z-plane



w-plane

4. (a) Evaluate $\int_c \operatorname{Re}(z) dz$ where c is a straight line from 0 to $1+2i$

(b) Show that $\int_0^{\infty} \frac{dx}{1+x^4} = \frac{\pi}{2\sqrt{2}}$

Ans: (a)

$$z = x+iy \Rightarrow \operatorname{Re} z = x$$

$$\therefore \int_c \operatorname{Re}(z) dz = \int_c x dz$$

c : straight line from 0 to $1+2i$

ie, $(0,0)$ to $(1,2)$

x varies from 0 to 1 & y varies from 0 to 2

$$\text{Also, } (1,2) \Rightarrow y = 2x$$

$$z = x+iy \Rightarrow dz = dx + i dy$$

$$dz = dx + i(2dx) = (1+2i) dx$$

$$\therefore \int_c x dz = \int_0^1 x(1+2i) dx = (1+2i) \left[\frac{x^2}{2} \right]_0^1 = \frac{1}{2}(1+2i) + c$$

$$(b) \quad 2 \int_0^{\infty} \frac{dx}{1+x^4} = \int_{-\infty}^{\infty} \frac{dx}{1+x^4}$$

$$\Rightarrow \frac{1}{2} \int_{-\infty}^{\infty} \frac{dx}{1+x^4} = \int_0^{\infty} \frac{dx}{1+x^4}$$

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^4} = \int_c \frac{dz}{1+z^4}$$

$$Dz = 0 \Rightarrow 1+z^4 = 0$$

$$\Rightarrow z = \pm \frac{1}{\sqrt{2}}(1+i), \pm \frac{1}{\sqrt{2}}(1-i)$$

ie, Roots are: $z = \frac{1}{\sqrt{2}}(1+i), \frac{1}{\sqrt{2}}(1-i), -\frac{1}{\sqrt{2}}(1+i), -\frac{1}{\sqrt{2}}(1-i)$

∴ points: $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$

$(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$

$(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})$

$(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})$

$(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$

$(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})$

$\bullet (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$

$\bullet (\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})$

$$1+z^4 = 0$$

$$\Rightarrow z = \pm \frac{1}{\sqrt{2}}(1+i), \pm \frac{1}{\sqrt{2}}(1-i)$$

2 points $\frac{1}{\sqrt{2}}(1+i)$ & $\frac{1}{\sqrt{2}}(1-i)$ lie above the real axis

9b a simple pole lying inside C

$$\therefore \int_{-\infty}^{\infty} \frac{dx}{1+x^4} = \int_C \frac{dz}{1+z^4} = 2\pi i [\text{SOR}]_{\text{(upper)}} + \pi i [\text{SOR}]_{\text{(on real axis)}}$$

$$\therefore \int_{-\infty}^{\infty} \frac{dx}{1+z^4} = 2\pi i \left[\text{Res } f(z)_{z=\frac{1}{\sqrt{2}}(1+i)} + \text{Res } f(z)_{z=\frac{1}{\sqrt{2}}(1-i)} \right]$$

$$\text{Res } f(z)_{z=\frac{1}{\sqrt{2}}(1+i)} = \frac{1}{4z^3} \Big|_{z=\frac{1}{\sqrt{2}}(1+i)} = \frac{1}{4 \left(\frac{1}{\sqrt{2}}\right)^3 (1+i)^3} = \frac{1}{4 \times \frac{1}{2\sqrt{2}} (2i-2)}$$

$$= \frac{1}{4 \times \frac{1}{2\sqrt{2}} 2(i-1)} = \frac{\sqrt{2}}{4(i-1)}$$

$$\text{Res } f(z)_{z=\frac{1}{\sqrt{2}}(1-i)} = \frac{1}{4z^3} \Big|_{z=\frac{1}{\sqrt{2}}(1-i)} = \frac{1}{4 \left(\frac{1}{\sqrt{2}}\right)^3 (1-i)^3} = \frac{1}{4 \times \frac{1}{2\sqrt{2}} (2i+2)}$$

$$= \frac{1}{4 \times \frac{1}{2\sqrt{2}} 2(i+1)} = \frac{\sqrt{2}}{4(i+1)}$$

$$\therefore \int_{-\infty}^{\infty} \frac{dx}{1+x^4} = 2\pi i \left[\frac{\sqrt{2}}{4(i-1)} + \frac{\sqrt{2}}{4(i+1)} \right] = 2\pi i \times \frac{\sqrt{2}}{4} \left[\frac{i+1}{(i+1)(i-1)} \right]$$

$$= 2\pi i \times \frac{\sqrt{2}}{4} \times \frac{2i}{i^2 - 1^2} = 2\pi i \times \frac{\sqrt{2}}{4} \times \frac{2i}{-2} = \pi i \times \frac{\sqrt{2}}{2} \times \frac{2i}{-2}$$

$$= \pi i \times \frac{\sqrt{2}}{2} = \frac{\pi}{\sqrt{2}}$$

$$\therefore \int_0^{\infty} \frac{dx}{1+x^4} = \frac{1}{2} \int_{-\infty}^{\infty} \frac{dx}{1+x^4} = \frac{1}{2} \times \frac{\pi}{\sqrt{2}} = \frac{\pi}{2\sqrt{2}}$$

5. (a) Integrate $\frac{z^2}{z^2-1}$ counter clockwise around the circle $|z-1-i| = \frac{\pi}{2}$

by Cauchy Integral formula

(b) Evaluate $\int_C \frac{z-23}{z^2+z-5} dz$ where C is $|z-2-i| = 3.5$ by Cauchy

Residue theorem

Ans: $\int_C \frac{z^2}{z^2-1} dz$ where $C: |z-1-i| = \frac{\pi}{2} = \frac{3.14}{2} = 1.75$

$$\int_C \frac{z^2}{z^2-1} dz = \int_C \frac{z^2}{(z+1)(z-1)} dz$$

$$Df=0 \Rightarrow (z+1)(z-1)=0 \Rightarrow z=1, -1$$

$$z=1, |z-1-i| = |1-1-i| = |-i| = 1 < 3.5 = \sqrt{2} = 1.41 < 3.5$$

$\therefore z=1$ lies inside C

$$z=-1, |z-1-i| = |-1-1-i| = |-2-i| = \sqrt{4+1} = \sqrt{5} = 2.23 > 1.75$$

$\therefore z=-1$ lies outside C

$$\text{By CIF, } \int_C \frac{f(z)}{z-a} dz = 2\pi i f(a)$$

$$\text{Thus, } \int_C \frac{z^2}{z^2-1} dz = \int_C \frac{z^2}{(z+1)(z-1)} dz = 2\pi i \times f(1) = 2\pi i \times \frac{1}{2} = \pi i$$

$$f(z) = \frac{z^2}{z+1}, \quad a=1$$

$$f(1) = \frac{1}{1+1} = \frac{1}{2}$$

(b) $\int_C \frac{z-23}{z^2-4z-5} dz$ where $C: |z-2-i| = 3.5$

here $f(z) = \frac{z-23}{z^2-4z-5}$

$D_f = 0 \Rightarrow z^2-4z-5=0 \Rightarrow z = \frac{4 \pm \sqrt{16+4(5)}}{2} = \frac{4 \pm \sqrt{36}}{2}$

$\Rightarrow z = \frac{4 \pm 6}{2} = 5, -1$

$z=5, |z-2-i| = |5-2-i| = |3-i| = \sqrt{10+1} = \sqrt{11} = 3.3 < 3.5$

$\therefore z=5$ lies inside

$z=-1, |z-2-i| = |-1-2-i| = |-3-i| = \sqrt{10+1} = \sqrt{11} = 3.3 < 3.5$

$\therefore z=-1$ lies inside

$\therefore z=5$ and $z=-1$ are simple poles of $f(z)$ lying inside C

By CRT, $\int_C f(z) dz = 2\pi i [\text{Sum of Residues}]$

$\Rightarrow \int_C \frac{z-23}{z^2-4z-5} dz = 2\pi i \left[\text{Res } f(z)_{z=5} + \text{Res } f(z)_{z=-1} \right]$

$\text{Res } f(z)_{z=5} = \frac{z-23}{z^2-4} \Big|_{z=5} = \frac{5-23}{10-4} = \frac{-18}{6} = -3$

$\text{Res } f(z)_{z=-1} = \frac{z-23}{z^2-4} \Big|_{z=-1} = \frac{-1-23}{-2-4} = \frac{-24}{-6} = 4$

$\therefore \int_C \frac{z-23}{z^2-4z-5} dz = 2\pi i [-3+4] = 2\pi i [1] = \underline{2\pi i}$

6(a) If $f(z) = \frac{1}{z^2}$, find the Taylor series that converges in $|z-i| < R$

& the Laurent's series that converges in $|z-i| > R$

(b) Define the 3 types of isolated singularities with an example for each.

Ans: (a) $|z-i| < R$

$f(z) = \frac{1}{z^2} = \frac{1}{(z-i+i)^2} = (i)^2 \frac{1}{\left[\frac{z-i}{i} + 1\right]^2}$
 $= - \left[\left(\frac{z-i}{i}\right) + 1 \right]^{-2}$

$$= - \left[1 - 2 \left(\frac{z-i}{i} \right) + 3 \left(\frac{z-i}{i} \right)^2 - 4 \left(\frac{z-i}{i} \right)^3 + \dots \right] \quad (x+i)^{-4} = 1 - 2x + 3x^2 - 4x^3 + \dots$$

$$\therefore f(z) = - \left[1 - 2 \left(\frac{z-i}{i} \right) + 3 \left(\frac{z-i}{i} \right)^2 - 4 \left(\frac{z-i}{i} \right)^3 + \dots \right]$$

when $|z-i| < R$

$$|z-i| > R$$

$$\begin{aligned} f(z) &= \frac{1}{z^2} = \frac{1}{(z-i+i)^2} = \frac{1}{(z-i)^2} \left[1 + \frac{i}{z-i} \right]^{-2} \\ &= \frac{1}{(z-i)^2} \left[1 - 2 \left(\frac{i}{z-i} \right) + 3 \left(\frac{i}{z-i} \right)^2 - 4 \left(\frac{i}{z-i} \right)^3 + \dots \right] \end{aligned}$$

$$\therefore f(z) = \frac{1}{(z-i)^2} - \frac{2i}{(z-i)^3} + \frac{3}{(z-i)^3} + \frac{4i}{(z-i)^4} - \dots$$

when $|z-i| > R$

(b) Refer NB, Pg no: 12, 13 (Module 5)

MODEL QUESTION PAPER MARCH 2017

1(a) Check whether the following functions are analytic or not. Justify your answer:

(i) $f(z) = z + \bar{z}$

(ii) $f(z) = |z|^2$

(b) Show that $f(z) = \sin z$ is analytic for all z . Find $f'(z)$

Ans: (a) (i) $f(z) = z + \bar{z}$

put $z = x+iy$ & $\bar{z} = x-iy$

$$f(z) = x+iy + x-iy = 2x.$$

$$f(z) = u+iv \Rightarrow u+iv = 2x.$$

$$\therefore u = 2x, v = 0.$$

$$u_x = 2, u_y = 0, v_x = 0, v_y = 0, \text{ continuous}$$

But $u_x \neq v_y$

C-R equations are not satisfied $\Rightarrow f(z) = z + \bar{z}$ is not analytic

$$(ii) f(z) = |z|^2$$

$$\text{Let } z = x+iy \Rightarrow |z| = \sqrt{x^2+y^2}, \Rightarrow |z|^2 = x^2+y^2$$

$$f(z) = u+iv$$

$$\Rightarrow u+iv = x^2+y^2$$

$$\therefore u = x^2+y^2 \quad \& \quad v = 0$$

$$u_x = 2x; \quad u_y = 2y; \quad v_x = 0; \quad v_y = 0, \text{ continuous}$$

$$u_x = v_y \Rightarrow 2x \neq 0$$

$$u_y = -v_x \Rightarrow 2y \neq 0$$

C-R equations will be satisfied only at $\underbrace{x=0, y=0}_{\text{i.e. } z=0}$

$\therefore f(z) = |z|^2$ is not analytic except at $z=0$

2.(a) Show that $v = 3x^2y - y^3$ is harmonic & find the corresponding analytic function $f(z) = u(x,y) + iv(x,y)$

(b) Find the image of $0 < x < 1, \frac{1}{2} < y < 1$ under the mapping $w = e^z$

Ans: (a) $v = 3x^2y - y^3$

$$v_x = 6xy$$

$$v_{xx} = 6y$$

$$v_{xy} = 6x$$

} all exist & are continuous

$$v_y = 3x^2 - 3y^2$$

$$v_{yy} = -6y$$

$$v_{yx} = 6x$$

$$\text{Also, } v_{xx} + v_{yy} = 6y - 6y = 0$$

$\therefore v = 3x^2y - y^3$ is harmonic

$f(z)$ is analytic $\Rightarrow f(z)$ satisfies the C-R equations

$$u_x = v_y \quad \& \quad u_y = -v_x$$

$$u_x = 3x^2 - 3y^2 \Rightarrow \frac{\partial u}{\partial x} = 3x^2 - 3y^2 \Rightarrow u = \int 3x^2 - 3y^2 dx$$

$$u = \frac{3x^3}{3} - 3xy^2 + g(y) \Rightarrow u = x^3 - 3xy^2 + g(y)$$

$$u_y = -v_x \Rightarrow u_y = -6xy \Rightarrow \frac{\partial u}{\partial y} = -6xy \Rightarrow \int -6xy dy$$

$$\Rightarrow -6xy + g'(y) = -6xy$$

$$\Rightarrow g'(y) = 0 \Rightarrow g(y) = \int 0 dy = 0 + C = C$$

$$\therefore f(z) = u(x,y) + i v(x,y) = x^3 - 3xy^2 + c + i [3x^2y - y^3]$$

(b)

$$w = e^z$$

$$\text{Let } z = x+iy \text{ \& } w = R e^{i\phi}$$

$$R e^{i\phi} = e^{x+iy} = e^x \cdot e^{iy}$$

$$R e^{i\phi} = e^x \cdot e^{iy}$$

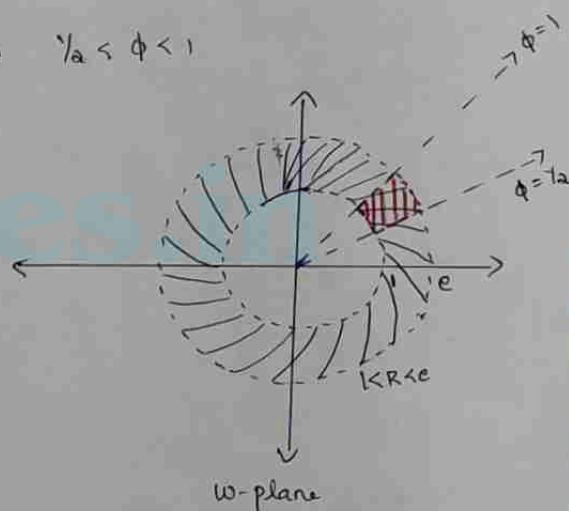
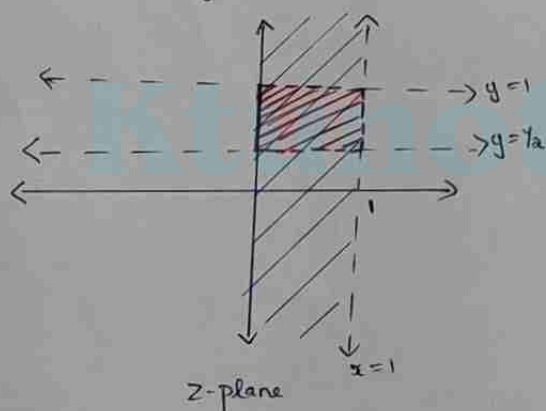
$$\text{Comparing, } R = e^x, \phi = y$$

$$0 < x < 1 \Rightarrow e^0 < e^x < e^1 \Rightarrow 1 < R < e$$

$\therefore 0 < x < 1$ will be mapped to the annulus $1 < R < e$

$$\frac{1}{2} < y < 1 \Rightarrow \frac{1}{2} < \phi < 1$$

$\therefore \frac{1}{2} < y < 1$ will be mapped to $\frac{1}{2} < \phi < 1$



1.(b)

$$f(z) = \sin z$$

$$z = x+iy$$

$$f(z) = \sin(x+iy) = \sin x \cosh y + i \cos x \sinh y$$

$$= \sin x \cosh y + i \cos x \sinh y$$

$$f(z) = u+iv$$

$$u+iv = \sin x \cosh y + i \cos x \sinh y$$

$$u = \sin x \cosh y$$

$$v = \cos x \sinh y$$

$$\sin(A+B) = \sin A \cosh B + \cos A \sinh B$$

$$\cosh y = \cosh y$$

$$\sinh y = i \cosh y$$

$$\frac{d}{dy} \cosh y = \sinh y$$

$$\frac{d}{dy} \sinh y = \cosh y$$

$$\left. \begin{aligned} u_x &= \cos x \cosh y & u_y &= \sin x \sinh y \\ v_x &= -\sin x \sinh y & v_y &= \cos x \cosh y \end{aligned} \right\} \text{continuous}$$

Also, $u_x = v_y$ & $u_y = -v_x$

∴ C-R equations are also satisfied

∴ $f(z) = \sin z$ is analytic

$$\begin{aligned} f'(z) &= u_x + i v_x \\ &= \cos x \cosh y + i [-\sin x \sinh y] \end{aligned}$$

If $f(z) = u + iv$ is analytic,
 $f'(z) = u_x + i v_x$

$$\cosh y = \cosh y$$

$$\sinh y = i \sinh y \Rightarrow \sinh y = \frac{\sinh y}{i}$$

$$\therefore f'(z) = \cos x \cosh y + i \left[-\sin x \frac{\sinh y}{i} \right]$$

$$f'(z) = \cos x \cosh y - \sin x \sinh y$$

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$f'(z) = \cos(x+iy)$$

$$\underline{\underline{f'(z) = \cos z}}$$

3. Find the image of the rectangular region $-\pi \leq x \leq \pi$, $a \leq y \leq b$ under the mapping $w = \sin z$

Ans:

$$w = \sin z$$

Let $w = u + iv$ & $z = x + iy$

$$\therefore u + iv = \sin(x + iy) = \sin x \cosh y + i \cos x \sinh y$$

$$u + iv = \sin x \cosh y + i \cos x \sinh y$$

$$\Rightarrow \left. \begin{aligned} u &= \sin x \cosh y \\ v &= \cos x \sinh y \end{aligned} \right\} \text{①}$$

Consider, $\cosh^2 y - \sinh^2 y = 1$

From ①, $\cosh y = \frac{u}{\sin x}$; $\sinh y = \frac{v}{\cos x}$

$$\therefore \frac{u^2}{\sin^2 x} - \frac{v^2}{\cos^2 x} = 1$$

$$\sinh^2 x + \cosh^2 x = 1$$

From (1), $\sinh x = \frac{u}{\cosh y}$ & $\cosh x = \frac{v}{\sinh y}$

$$\therefore \frac{u^2}{\cosh^2 y} + \frac{v^2}{\sinh^2 y} = 1$$

† (a) Evaluate $\int_C |z| dz$ where

(i) C is the line segment joining $-i$ & i

(ii) C is the unit circle in the left of half plane

(b) Verify Cauchy's Integral theorem for z^2 taken over the boundary of the rectangle with vertices $-1, 1, 1+i, -1+i$ in the counter clockwise direction.

Ans: (a) (i) $\int_C |z| dz$ where C is the line segment from $-i$ to i

$$z = x + iy \Rightarrow |z| = \sqrt{x^2 + y^2}$$

C is the line segment from $-i$ to i

i.e., $(0, -1)$ to $(0, 1)$.

here $x = 0$.

$$\Rightarrow |z| = \sqrt{0 + y^2} = y$$

$$z = x + iy = iy \Rightarrow dz = i dy$$

y varies from -1 to 1

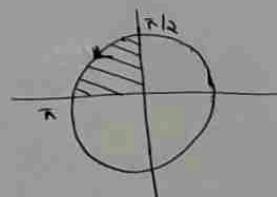
$$\therefore \int_C |z| dz = \int_{-1}^1 y i dy = i \int_{-1}^1 y dy = i \left[\frac{y^2}{2} \right]_{-1}^1$$

$$= i \left[\frac{1}{2} - \frac{1}{2} \right] = \underline{\underline{0}}$$

(ii) By Second evaluation method,

$$\int_C f(z) dz = \int_a^b f(z(\theta)) \dot{z}(\theta) d\theta$$

where $C: z(\theta)$, $a \leq \theta \leq b$.



here $f(z) = |z|$

c: $z(\theta) = e^{i\theta}$, $\pi/2 \leq \theta \leq \pi$

$f(z(\theta)) = f(e^{i\theta}) = |e^{i\theta}| = 1$

$\dot{z}(\theta) = \frac{dz}{d\theta} = \frac{d}{d\theta}(e^{i\theta}) = ie^{i\theta}$

$\therefore \int_c |z| dz = \int_a^b f(z(\theta)) \dot{z}(\theta) d\theta = \int_{\pi/2}^{\pi} ie^{i\theta} d\theta = i \left[\frac{e^{i\theta}}{i} \right]_{\pi/2}^{\pi}$

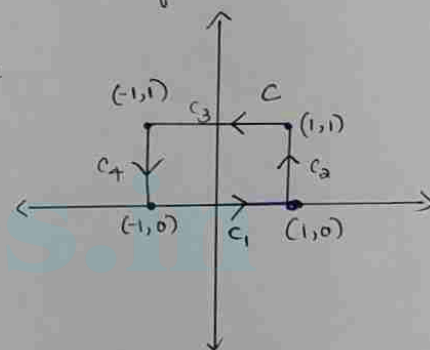
$= e^{i\pi} - e^{i\pi/2} = (\cos \pi + i \sin \pi) - (\cos \pi/2 + i \sin \pi/2)$

$= (-1 + 0) - (0 + i) = -1 - i$

Ans: (b)

$\int_c z^2 dz = 0$ by Cauchy Integral theorem ($\because z^2$ is an analytic function)

$\int_c z^2 dz = \int_{c_1} z^2 dz + \int_{c_2} z^2 dz + \int_{c_3} z^2 dz + \int_{c_4} z^2 dz$



Along c_1 : $c_1 = x$ -axis

$c_1: y = 0.$

$z^2 = (x+iy)^2 = x^2$

$z = x+iy = x \Rightarrow dz = dx.$

x varies from -1 to 1

$\therefore \int_{c_1} z^2 dz = \int_{-1}^1 x^2 dx = \left[\frac{x^3}{3} \right]_{-1}^1 = \frac{1}{3} - \left(\frac{-1}{3} \right) = \frac{2}{3}$

Along c_2 : c_2 varies from $(1,0)$ to $(1,1)$

$\therefore c_2: x = 1$

$z^2 = (x+iy)^2 = (1+iy)^2 = 1 + 2iy - y^2$

$z = x+iy = 1+iy \Rightarrow dz = i dy$

y varies from 0 to 1

$\therefore \int_{c_2} z^2 dz = \int_0^1 (1 + 2iy - y^2) i dy = i \left[y + \frac{2iy^2}{2} - \frac{y^3}{3} \right]_0^1 = i \left[1 + i - \frac{1}{3} \right] = \frac{2i}{3} - i$

Along C_3 : C_3 varies from $(1,1)$ to $(-1,1)$

$$\therefore C_3 : y=1$$

$$z^2 = (x+iy)^2 = (x+i)^2 = x^2 + 2ix - 1$$

$$z = x+iy = x+i \Rightarrow dz = dx$$

x varies from 1 to -1

$$\begin{aligned} \therefore \int_{C_3} z^2 dz &= \int_1^{-1} (x^2 + 2ix - 1) dx = \left[\frac{x^3}{3} + ix^2 - x \right]_1^{-1} \\ &= \frac{-1}{3} + i + 1 - \frac{1}{3} - i + 1 = -\frac{2}{3} + 2 = \frac{4}{3} \end{aligned}$$

Along C_4 : ~~C_4 varies from~~

C_4 varies from $(-1,1)$ to $(-1,0)$

$$\therefore x = -1$$

$$z^2 = (x+iy)^2 = (-1+iy)^2 = 1 - 2iy - y^2$$

$$z = x+iy = -1+iy \Rightarrow dz = i dy$$

y varies from 1 to 0

$$\begin{aligned} \therefore \int_{C_4} z^2 dz &= \int_1^0 (1 - 2iy - y^2) i dy = i \left[y - iy^2 - \frac{y^3}{3} \right]_1^0 \\ &= 0 - i \left[1 - i - \frac{1}{3} \right] = -i \left[\frac{2}{3} - i \right] = -\frac{2i}{3} - 1 \end{aligned}$$

$$\begin{aligned} \therefore \int_C z^2 dz &= \int_{C_1} z^2 dz + \int_{C_2} z^2 dz + \int_{C_3} z^2 dz + \int_{C_4} z^2 dz \\ &= \frac{2}{3} + \frac{2i}{3} - 1 + \frac{4}{3} - \frac{2i}{3} - 1 = \frac{6}{3} - 2 = 2 - 2 = \underline{\underline{0}} \end{aligned}$$

Hence Cauchy Integral theorem is verified

5 (a) Find the Laurent's series expansion of $f(z) = \frac{1}{1-z^2}$ which is convergent in : (i) $|z-1| < 2$ (ii) $|z-1| > 2$

(b) Determine the nature & type of singularities of :

(i) $\frac{e^{-z^2}}{z^2}$

(ii) $z \sin\left(\frac{1}{z}\right)$

Ans: (a) (i) $f(z) = \frac{1}{1-z^2} = \frac{1}{(1+z)(1-z)} = \frac{A}{1+z} + \frac{B}{1-z}$

$$1 = A(1-z) + B(1+z)$$

$$z=1, \quad 1 = 2B \Rightarrow B = \frac{1}{2}$$

$$z=-1, \quad 1 = 2A \Rightarrow A = \frac{1}{2}$$

$$\therefore \frac{1}{1-z^2} = \frac{1}{2(1+z)} + \frac{1}{2(1-z)}$$

$$= \frac{1}{2(1+z+1)} + \frac{1}{-2(z-1)} = \frac{1}{2(z-1+2)} - \frac{1}{2(z-1)}$$

$$= \frac{1}{2\left(1+\frac{z-1}{2}\right)} - \frac{1}{2(z-1)}$$

$$= \frac{1}{2} \left[1 + \left(\frac{z-1}{2}\right) \right]^{-1} + \frac{1}{2} (z-1)^{-1}$$

$$f(z) = \frac{1}{2} \left[1 - \left(\frac{z-1}{2}\right) + \left(\frac{z-1}{2}\right)^2 - \dots \right] + \frac{1}{2} (1+z+z^2+z^3+\dots)$$

(ii) $f(z) = \frac{1}{1-z^2} = \frac{1}{(1+z)(1-z)} = \frac{A}{1+z} + \frac{B}{1-z}$

$$f(z) = \frac{1}{2(1+z)} + \frac{1}{2(1-z)} = \frac{1}{2(z-1+1)} - \frac{1}{2(z-1)}$$

$$= \frac{1}{2(z-1+2)} - \frac{1}{2(z-1)} = \frac{1}{2(z-1)\left[1+\frac{2}{z-1}\right]} - \frac{1}{2(z-1)}$$

$$= \frac{1}{2(z-1)} \left[1 + \frac{2}{z-1} \right]^{-1} - \frac{1}{2} (z-1)^{-1}$$

$$= \frac{1}{2(z-1)} \left[1 - \frac{2}{z-1} + \left(\frac{2}{z-1}\right)^2 - \dots \right] + \frac{1}{2} (1+z+z^2+\dots)$$

(b) (i) $\frac{e^{-z^2}}{z^2} = \frac{1}{z^2} \left[1 - z^2 + \frac{z^4}{2!} - \frac{z^6}{3!} + \dots \right]$

$$\frac{e^{-z^2}}{z^2} = \frac{1}{z^2} - 1 + \frac{z^2}{2!} - \frac{z^4}{3!} + \dots$$

$$\boxed{e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots}$$

Since the is only a finite principal part

$z=0$ is a pole of order 2

(ii) $z \sin\left(\frac{1}{z}\right) = z \left[\frac{1}{z} - \frac{1}{z^3 \cdot 3!} + \frac{1}{z^5 \cdot 5!} - \dots \right]$

$$z \sin\left(\frac{1}{z}\right) = 1 - \frac{1}{z^2 \cdot 3!} + \frac{1}{z^4 \cdot 5!} - \dots$$

$$\boxed{\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots}$$

Since there is infinite principal part, $z=0$ is an essential singularity

6. (a) Use residue theorem to evaluate $\int_C \frac{30z^2 - 23z + 5}{(2z-1)^2(3z-1)} dz$ where

C is $|z| = 1$

(b) Evaluate $\int_0^{\infty} \frac{1}{(1+x)^2} dx$ using residue theorem

Ans: (a) $f(z) = \frac{30z^2 - 23z + 5}{(2z-1)^2(3z-1)}$

$$Df=0 \Rightarrow (2z-1)^2(3z-1)=0 \Rightarrow (2z-1)^2=0 \text{ or } 3z-1=0$$

$$\Rightarrow z = \frac{1}{2}, \frac{1}{3}$$

$z = \frac{1}{2}$ is a pole of order 2 & $z = \frac{1}{3}$ is a simple pole.

By CRT, $\int_C f(z) dz = 2\pi i \left[\text{Sum of residues} \right]$

$$\therefore \int_C \frac{30z^2 - 23z + 5}{(2z-1)^2(3z-1)} dz = 2\pi i \left[\text{Res } f(z)_{z=\frac{1}{2}} + \text{Res } f(z)_{z=\frac{1}{3}} \right]$$

$$\text{Res } f(z)_{z=\frac{1}{2}} = \lim_{z \rightarrow \frac{1}{2}} \frac{1}{(2z-1)!} \frac{d^{2-1}}{dz^{2-1}} \left[(z-\frac{1}{2})^2 f(z) \right]$$

$$= \lim_{z \rightarrow \frac{1}{2}} \frac{d}{dz} \left[\frac{(2z-1)^2}{4} \cdot \frac{30z^2 - 23z + 5}{(2z-1)^2(3z-1)} \right]$$

$$= \lim_{z \rightarrow \frac{1}{2}} \frac{1}{4} \frac{d}{dz} \left[\frac{30z^2 - 23z + 5}{3z-1} \right]$$

$$= \frac{1}{4} \lim_{z \rightarrow \frac{1}{2}} \left[\frac{(3z-1)(60z-23) - (30z^2 - 23z + 5) \cdot 3}{(3z-1)^2} \right]$$

$$= \frac{1}{4} \left[\frac{(\frac{3}{2}-1)(30-23) - (30(\frac{1}{2})^2 - 23(\frac{1}{2}) + 5) \cdot 3}{(\frac{3}{2}-1)^2} \right]$$

$$= \frac{1}{2}$$

~~Pole of order m~~
 ~~$\text{Res } f(z)_{z=a} = \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}} [(z-a)^m f(z)]$~~

Pole of order m
 $\text{Res } f(z)_{z=a} = \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}} [(z-a)^m f(z)]$
 as $\lim_{z \rightarrow a}$

$$\text{Res } f(z)_{z=\frac{1}{3}} = \lim_{z \rightarrow \frac{1}{3}} (z-\frac{1}{3}) \frac{30z^2 - 23z + 5}{(2z-1)^2(3z-1)} = \frac{1}{3} \lim_{z \rightarrow \frac{1}{3}} \frac{(2z-1)(30z^2 - 23z + 5)}{(2z-1)^2(3z-1)}$$

$$= \frac{1}{3} \left[\frac{30 \times \frac{1}{9} - 23 \times \frac{1}{3} + 5}{(2 \times \frac{1}{3} - 1)^2} \right] = 2$$

$$\therefore \int_0^{\infty} \frac{30z^2 - 232z + 5}{(z^2-1)^2(z^2-1)} dz = 2\pi i \left[\frac{1}{2} + 2 \right] = 5\pi i$$

(b) Refer NB, Module 5, Qno: 3, pg no: 49

MODEL QUESTION PAPER

JULY 2017

1. Does the limit $\lim_{z \rightarrow 0} \frac{z}{\bar{z}}$ exist? If yes, find the value. If no, explain why?

Ans: Let $z = x+iy$, $\bar{z} = x-iy$

$$z \rightarrow 0 \Rightarrow (x,y) \rightarrow 0 \Rightarrow x \rightarrow 0; y \rightarrow 0.$$

$$\lim_{\substack{z \rightarrow 0 \\ y \rightarrow 0}} \frac{z}{\bar{z}} = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x+iy}{x-iy} = \lim_{x \rightarrow 0} \frac{x+imx}{x-imx} = \lim_{x \rightarrow 0} \frac{(1+im)x}{(1-im)x}$$

$$\text{put } y = mx$$

$$\lim_{z \rightarrow 0} \frac{z}{\bar{z}} = \lim_{x \rightarrow 0} \frac{1+im}{1-im} = \frac{1+im}{1-im}, \text{ infinite no. of solutions for different values of } m$$

$$\therefore \lim_{z \rightarrow 0} \frac{z}{\bar{z}} \text{ does not exist}$$

2(a) Find the points, if any, in complex plane where the function $f(z) = 2x^2 + y + i(y^2 - x)$ is (i) differentiable (ii) analytic

Ans: $f(z) = 2x^2 + y + i(y^2 - x)$

$$\text{here } u = 2x^2 + y \quad \& \quad v = y^2 - x$$

$$u_x = 4x$$

$$u_y = 1$$

$$v_x = -1$$

$$v_y = 2y$$

Partial derivatives are continuous but CR equations are not satisfied

$$\text{But } u_x = v_y \text{ at } x=0, y=0$$

$$u_y = -v_x$$

$\therefore f(z)$ is analytic only at $(0,0)$ i.e; its differentiable at $(0,0)$

(b) Show that the function $u(x,y) = x^3 - 3xy^2 - 5y$ is harmonic everywhere. Also find the harmonic conjugate v .

Ans.

$$u_x = 3x^2 - 3y^2$$

$$u_y = -6xy - 5$$

$$u_{xy} = -6y$$

$$u_{xx} = 6x$$

$$u_{yy} = -6x$$

$$u_{yx} = -6y$$

They are continuous everywhere.

$$\text{Also, } u_{xx} + u_{yy} = 6x - 6x = 0.$$

$\therefore u(x,y) = x^3 - 3xy^2 - 5y$ is harmonic

For harmonic conjugate, refer module 3, pg no: 38

3(a) Evaluate $\int_c \bar{z} dz$ where c is given by $x = 3t, y = t^2, -1 \leq t \leq 1$

(b) Show that $\int_c (z+z)^2 dz = \frac{-i}{3}$ where c is any path connecting -2 and $-2+i$

Ans (a)

$$z = x+iy \Rightarrow \bar{z} = x-iy$$

$$x=3t, y=t^2 \Rightarrow \bar{z} = 3t - it^2, -1 \leq t \leq 1$$

$$z = x+iy = 3t + it^2$$

$$dz = 3dt + 2it dt$$

$$= (3+2it)dt$$

$$\therefore \int_c \bar{z} dz = \int_{-1}^1 (3t - it^2)(3+2it) dt = \int_{-1}^1 (9t + 6it^2 - 3it^2 + 2t^3) dt$$

$$= \int_{-1}^1 (9t + 3it^2 + 2t^3) dt = \left[\frac{9t^2}{2} + it^3 + \frac{t^4}{2} \right]_{-1}^1$$

$$= (72 + 64i + 128) - \left(\frac{9}{2} - i + \frac{1}{2} \right) = \underline{\underline{195 + 65i}}$$

(b) $-2 \rightarrow (-2,0) \Rightarrow x = -2$

$-2+i \rightarrow (-2,1)$

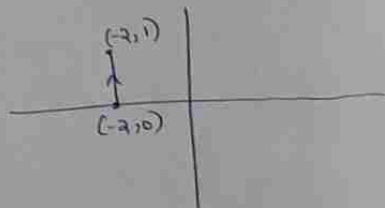
$$(z+z)^2 = (z+x+iy)^2 = (z-2+iy)^2 = -y^2$$

$$z = x+iy = -2+iy$$

$$\Rightarrow dz = idy$$

y varies from 0 to 1

$$\therefore \int_c (z+z)^2 dz = i \int_0^1 -y^2 dy = -i \left[\frac{y^3}{3} \right]_0^1 = \underline{\underline{\frac{-i}{3}}}$$



4. (a) Evaluate $\int_C \frac{5z+7}{z^2+2z-3} dz$ where C is the circle $|z-2|=2$.

(b) Find the Laurent's series expansion of $\frac{1}{z-2}$ in $1 < |z+1| < 2$.

Ans: (a) $D\tau=0 \Rightarrow z^2+2z-3=0 \Rightarrow z = \frac{-2 \pm \sqrt{4+12}}{2} = \frac{-2 \pm \sqrt{16}}{2} = \frac{-2 \pm 4}{2}$

$\Rightarrow z = 1, -3$

$\therefore z^2+2z-3 = (z-1)(z-(-3)) = (z-1)(z+3)$

$C: |z-2|=2$.

$z=1, |z-2|=|1-2|=|-1|=1 < 2$, lies inside C

$z=-3, |z-2|=|-3-2|=|-5|=5 > 2$, lies outside C

$\therefore \int_C \frac{5z+7}{z^2+2z-3} dz = \int_C \frac{f(z)}{(z-1)(z+3)} dz = 2\pi i f(1) = 2\pi i \times 3 = 6\pi i$

here $f(z) = \frac{5z+7}{z+3}$

$a=1$

$\therefore f(a) = f(1) = \frac{5+7}{1+3} = \frac{12}{4} = 3$

(b) Refer NB, Module 5, pg no: 5

5. (a) Use Cauchy Integral formula to evaluate $\int_C \frac{z+1}{z^4+2iz^3} dz$ where C is $|z|=1$

(b) Use contour integration, evaluate $\int_{-\infty}^{\infty} \frac{x^3-x+2}{x^4+10x^2+9} dx$

Ans: (a) $\int_C \frac{z+1}{z^4+2iz^3} dz = \int_C \frac{z+1}{z^3(z+2i)} dz$

$D\tau=0 \Rightarrow z^3(z+2i)=0 \Rightarrow z=0, -2i$

$C: |z|=1$

$z=0, |z|=|0|=0 < 1$, lies inside C

$z=-2i, |z|=|-2i|=2 > 1$, lies outside C

$\therefore \int_C \frac{z+1}{z^4+2iz^3} dz = \int_C \frac{f(z)}{z^3(z+2i)} dz = \frac{2\pi i f'(0)}{2!} = \frac{2\pi i \times 1}{2} = \pi i$

here $f(z) = \frac{z+1}{z+2i}$ & $a=0 \Rightarrow f(a) = f(0) = \frac{0+1}{0+2i} = \frac{1}{2i}$

$$f(z) = \frac{z+1}{z+2i} \quad \& \quad a=0$$

$$f'(z) = \frac{(z+2i) \cdot 1 - (z+1) \cdot i}{(z+2i)^2} = \frac{z+2i-z-i}{(z+2i)^2} = \frac{2i-1}{(z+2i)^2}$$

$$f''(z) = (2i-1) \left[\frac{-2}{(z+2i)^3} \right]$$

$$f''(0) = (2i-1) \times \frac{-2}{(2i)^3} = \frac{-2(2i-1)}{8(-i)} = \frac{2i-1}{4}$$

$$\therefore \int_C \frac{z+1}{z^2+2iz+3} dz = \frac{2\pi i}{2!} f''(0) = \pi i \times \frac{2i-1}{4} = \frac{(2i-1)\pi i}{4}$$

$$(b) \int_{-\infty}^{\infty} \frac{x^2-x+2}{x^4+10x^2+9} dx = \int_{-\infty}^{\infty} \frac{x^2-x+2}{x^4+9x^2+x^2+9} dx = \int_{-\infty}^{\infty} \frac{x^2-x+2}{x^2(x^2+9)+1(x^2+9)} dx$$

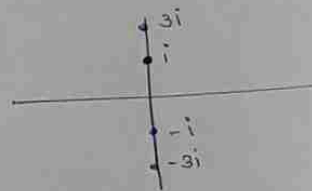
$$= \int_{-\infty}^{\infty} \frac{x^2-x+2}{(x^2+1)(x^2+9)} dx$$

Consider $\int_{-\infty}^{\infty} \frac{z^2-z+2}{(z^2+1)(z^2+9)} dz$

$$D_0 = 0 \Rightarrow (z^2+1)(z^2+9) = 0 \Rightarrow z^2 = -1 \text{ or } z^2 = -9$$

$$\Rightarrow z = \pm i \text{ or } z = \pm 3i$$

$z = i$ & $z = 3i$ lies above x -axis



By CRT,

$$\int_{-\infty}^{\infty} \frac{z^2-z+2}{(z^2+1)(z^2+9)} dz = 2\pi i \left[\text{Sum of residues (above } x\text{-axis)} \right] + \pi i \left[\text{Sum of residues (on } x\text{-axis)} \right]$$

$z=i$

It is a simple pole.

$$\therefore \text{Res } f(z) = \lim_{z \rightarrow i} \frac{z^2-z+2}{(z^2+1)2z+(z^2+9)2z} \Big|_{z=i} = \frac{-1-i+2}{0+2i \times 8} = \frac{1-i}{16i}$$

$z=3i$

It is a

$$\therefore \text{Res } f(z) = \frac{z - z + a}{(z^2 + 1)2z + (z^2 + 9)2z} \Big|_{z=3i} = \frac{-9 - 3i + a}{-8 \times 6i + 0} = \frac{-7 - 3i}{-48i}$$

$$\begin{aligned} \therefore \int_{-\infty}^{\infty} \frac{x^2 - x + 2}{(x^2 + 1)(x^2 + 9)} dx &= \int_{-\infty}^{\infty} \frac{z^2 - z + 2}{(z^2 + 1)(z^2 + 9)} dz \\ &= 2\pi i \left[\frac{1-i}{16i} + \frac{-7-3i}{-48i} \right] + \pi i [0] \\ &= 2\pi i \left[\frac{3(1-i) + (7+3i)}{48} \right] = \frac{\pi}{24} [3 - 3i + 7 + 3i] \\ &= \frac{\pi}{24} \times (10) = \underline{\underline{\frac{5\pi}{12}}} \end{aligned}$$

MODEL QUESTION PAPER

COMPLEX ANALYSIS

DECEMBER 2017

1. (a) Find the points where Cauchy - Riemann equations are satisfied for the function $f(z) = xy^2 + ix^3y$. Where does $f'(z)$ exist? Is the function $f(z)$ analytic at those points?

(b) Show that $u = x^3 - y^3 - y$ is harmonic. Also find the corresponding conjugate harmonic function.

Ans: (a)

$$f(z) = xy^2 + ix^3y$$

$$u = xy^2$$

$$v = x^3y$$

$$u_x = y^2$$

$$v_x = 3x^2y$$

$$u_{xx} = 0$$

$$v_{xx} = 6xy$$

(b) (a)

$$u_y = 2xy$$

$$v_y = x^3$$

$$u_{yy} = 2x$$

$$v_{yy} = 0$$

$$u_x = v_y$$

only at $x=0, y=0$

$$u_y = -v_x$$

ie, only at $z=0$.

$\therefore$ CR equations will be satisfied only at $z=0$ for $f(z)$

$\therefore f'(z)$ exist at that point & $f(z)$ is analytic at that point.

2. (a) If $v = e^x (x \sin y + y \cos y)$, find an analytic function $f(z) = u + iv$.
 (b) Test the continuity at $z=0$ if $f(z) = \begin{cases} \frac{\operatorname{Im} z}{|z|}, & z \neq 0 \\ 0, & z = 0 \end{cases}$

Ans: (a) $v = e^x (x \sin y + y \cos y) = x e^x \sin y + y e^x \cos y$

If $f(z)$ is an analytic function, u & v satisfies the C-R equation

$$U_x = V_y \quad \& \quad U_y = -V_x$$

$$U_x = V_y \Rightarrow U_x = x e^x \cos y + e^x [-y \sin y + \cos y]$$

$$\text{ie, } U_x = x e^x \cos y - y e^x \sin y + e^x \cos y$$

$$\text{ie, } \frac{\partial u}{\partial x} = x e^x \cos y - y e^x \sin y + e^x \cos y$$

$$u = \int (x e^x \cos y - y e^x \sin y + e^x \cos y) dx$$

$$u = \cos y \int x e^x dx - y \sin y \int e^x dx + \cos y \int e^x dx + g(y)$$

$$\text{ie, } u = \cos y [x e^x - e^x] - y \sin y e^x + e^x \cos y + g(y)$$

$$u = x e^x \cos y - e^x \cos y - y e^x \sin y + e^x \cos y + g(y)$$

$$\therefore u = x e^x \cos y - y e^x \sin y + g(y)$$

$$U_y = -V_x$$

$$\Rightarrow -x e^x \sin y - e^x [y \cos y + \sin y] + g'(y)$$

$$= -[\sin y (x e^x + e^x)] + y \cos y e^x$$

$$-x e^x \sin y - y e^x \cos y - e^x \sin y + f'(y) = -e^x \sin y - x e^x \sin y - y e^x \cos y$$

$$\Rightarrow f'(y) = 0$$

$$\Rightarrow f(y) = c$$

$$\therefore u = x e^x \cos y - y e^x \sin y + g(y)$$

$$u = e^x (x \cos y - y \sin y) + c$$

$$\therefore f(z) = u + iv = \underline{e^x (x \cos y - y \sin y) + c + i e^x (x \sin y + y \cos y)}$$

$$(b) \quad f(z) = \begin{cases} \frac{\operatorname{Im} z}{|z|}, & z \neq 0 \\ 0, & z = 0 \end{cases}$$

$$\text{Let } z = x+iy \Rightarrow \operatorname{Im} z = y, \quad |z| = \sqrt{x^2 + y^2}.$$

$\therefore f(z) \leftarrow f(z)$ is said to be continuous at $z=0$ if,

$$\lim_{z \rightarrow 0} f(z) = f(0).$$

$$\lim_{z \rightarrow 0} f(z) = \lim_{\substack{x+iy \rightarrow 0 \\ x \rightarrow 0 \\ y \rightarrow 0}} f(z) = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{y}{\sqrt{x^2 + y^2}}$$

$$\text{put } y = mx.$$

$$\therefore \lim_{z \rightarrow 0} f(z) = \lim_{x \rightarrow 0} \frac{mx}{\sqrt{x^2 + m^2 x^2}} = \lim_{x \rightarrow 0} \frac{mx}{x \sqrt{1+m^2}} = \frac{m}{\sqrt{1+m^2}}$$

$\therefore$ Infinitely many solution exists

$\therefore$ limit does not exist

$\therefore f(z)$ is not continuous at $z=0$

3. Find the image of the line $x=1, y=2$ & $x>0, y<0$ under the mapping $w = z^2$.

Ans:

$$w = z^2$$

$$\text{put } w = u+iv \text{ \& } z = x+iy$$

$$\Rightarrow u+iv = (x+iy)^2 = x^2 + i2xy - y^2 = x^2 - y^2 + i2xy$$

$$\Rightarrow u = x^2 - y^2; \quad v = 2xy$$

$$x=1 \Rightarrow u = 1 - y^2 \quad \& \quad v = 2y \Rightarrow y = \frac{v}{2}.$$

$$y=2 \Rightarrow v = 4x \Rightarrow x = \frac{v}{4} \Rightarrow u = 1 - \frac{v^2}{4} \Rightarrow v^2 = -4(u-1)$$

$$\Rightarrow (v-0)^2 = -4(u-1), \text{ a parabola}$$

with vertex $(1,0)$ & open left

$$y=2 \Rightarrow u = x^2 - 4; \quad v = 4x \Rightarrow x = \frac{v}{4}$$

$$\therefore u = \frac{v^2}{16} - 4 \Rightarrow v^2 = 16(u+4) \Rightarrow (v-0)^2 = 16(u-(-4)),$$

a parabola with vertex $(-4,0)$ & opens right.

4. (a) Evaluate $\int_C \operatorname{Re} z^3 dz$ over the boundary c of the square with vertices $0, i, 1+i, 1$ clockwise

(b) Evaluate $\int \frac{4-3z}{z(z-1)}$ over the circle $|z| = \frac{3}{2}$

(c) Evaluate $\int \frac{3z^2+7z+1}{z+1} dz$ over the circle $|z+i|=1$

Ans: (a) $z = x+iy \Rightarrow z^3 = (x+iy)^3 = x^3 - y^3 + i(3xy^2 + x^2y)$

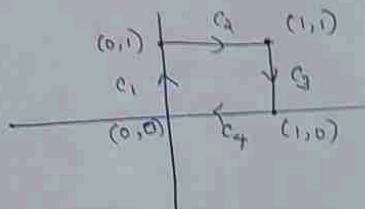
$$\operatorname{Re} z^3 = x^3 - y^3$$

vertices $0 \rightarrow (0,0)$

$i \rightarrow (0,1)$

$1+i \rightarrow (1,1)$

$1 \rightarrow (1,0)$



$$\int_C \operatorname{Re} z^3 dz = \int_{c_1} \operatorname{Re} z^3 dz + \int_{c_2} \operatorname{Re} z^3 dz + \int_{c_3} \operatorname{Re} z^3 dz + \int_{c_4} \operatorname{Re} z^3 dz$$

Along c_1 : $(0,0) \rightarrow (0,1)$

$$x = 0$$

$$\therefore \operatorname{Re} z^3 = x^3 - y^3 = -y^3$$

$$z = x+iy \Rightarrow z = iy \Rightarrow dz = i dy$$

y varies from 0 to 1

$$\therefore \int_{c_1} \operatorname{Re} z^3 dz = \int_0^1 -y^3 i dy = -i \left[\frac{y^4}{4} \right]_0^1 = -\frac{i}{4}$$

Along c_2 : $(0,1) \rightarrow (1,1)$

$$y = 1$$

$$\therefore \operatorname{Re} z^3 = x^3 - y^3 = x^3 - 1$$

$$z = x+iy \Rightarrow z = x+i \Rightarrow dz = dx$$

x varies from 0 to 1

$$\therefore \int_{c_2} \operatorname{Re} z^3 dz = \int_0^1 (x^3 - 1) dx = \left[\frac{x^4}{4} - x \right]_0^1 = \frac{1}{4} - 1 = -\frac{3}{4}$$

Along c_3 : $(1,1) \rightarrow (1,0)$

$$\because \text{Re } z^2 = x^2 - y^2 = 1 - y^2$$

$$z = x + iy \Rightarrow z = 1 + iy \Rightarrow dz = i dy$$

y varies from 1 to 0.

$$\therefore \int_{C_3} \text{Re } z^2 dz = \int_1^0 (1 - y^2) i dy = i \left[y - \frac{y^3}{3} \right]_1^0 = -\frac{2i}{3}$$

Along C_4 : $(1, 0) \rightarrow (0, 0)$

$$y = 0$$

$$\therefore \text{Re } z^2 = x^2 - y^2 = x^2$$

$$z = x + iy = x \Rightarrow dz = dx$$

x varies from 1 to 0.

$$\therefore \int_{C_4} \text{Re } z^2 dz = \int_1^0 x^2 dx = \left[\frac{x^3}{3} \right]_1^0 = -\frac{1}{3}$$

$$\therefore \int_C \text{Re } z^2 dz = \frac{-i}{3} - \frac{2}{3} - \frac{2i}{3} - \frac{1}{3} = \underline{\underline{-1-i}}$$

(b) $\int_C \frac{4-3z}{z(z-1)} dz$ where $C: |z| = \frac{3}{2}$

$$Df=0 \Rightarrow z(z-1)=0 \Rightarrow z=0, 1$$

$$z=0, |z|=|0|=0 < \frac{3}{2} ; z=0 \text{ lies inside } C$$

$$z=1, |z|=|1|=1 < \frac{3}{2} ; z=1 \text{ lies inside } C$$

Construct 2 circles C_1 & C_2 with centers 0 & 1 respectively such that C_1 & C_2 do not intersect & lies completely inside C

By Cauchy Integral formula by multiply connected domains,

$$\int_C \frac{4-3z}{z(z-1)} dz = \int_{C_1} \frac{4-3z}{z(z-1)} dz + \int_{C_2} \frac{4-3z}{z(z-1)} dz$$

$$= 2\pi i f_1(0) + 2\pi i f_2(1)$$

$$= 2\pi i \left[\frac{4-0}{0-1} \right] + 2\pi i \left[\frac{4-3}{1} \right]$$

$$= 2\pi i (-4) + 2\pi i = \underline{\underline{-6\pi i}}$$

$$(c) \int_0^{\infty} \frac{3z^2 + 7z + 1}{z+1} dz \quad \text{where } C: |z+i|=1$$

$$D=0 \Rightarrow z+i=0 \Rightarrow z=-i$$

$$\text{When } z=-i, |z+i| = |-i+i| = \sqrt{(-i)^2 + (i)^2} = \sqrt{2} \\ = 1.4 > 1, \text{ lies outside } C$$

$$\therefore \text{By Cauchy Integral theorem, } \int_C \frac{3z^2 + 7z + 1}{z+1} dz = 0$$

$$|z| = |x+iy| = \sqrt{x^2+y^2}$$

$$5(a) \text{ Expand } \frac{z}{(z-1)(z-2)} \text{ in (i) } 0 < |z-2| < 1 \quad (\text{ii}) |z-1| > 1$$

$$(b) \text{ Evaluate } \int_0^{2\pi} \frac{1}{2 + \cos \theta} d\theta$$

$$\text{Ans (a)} \quad \frac{z}{(z-1)(z-2)} = \frac{A}{z-1} + \frac{B}{z-2}$$

$$\Rightarrow z = A(z-2) + B(z-1)$$

$$z=2, \quad 2=B; \quad z=1, \quad 1=-A \Rightarrow A=-1$$

$$\therefore \frac{z}{(z-1)(z-2)} = \frac{-1}{z-1} + \frac{2}{z-2}$$

$$(i) \quad 0 < |z-2| < 1 \\ \frac{z}{(z-1)(z-2)} = \frac{-1}{z-2+2-1} + \frac{2}{z-2} = \frac{-1}{(z-2)+1} + \frac{2}{z-2} \\ = -[1+(z-2)]^{-1} + \frac{2}{z-2} \\ = -[1-(z-2) + (z-2)^2 - (z-2)^3 + \dots] + \frac{2}{z-2}$$

$$(ii) \quad |z-1| > 1 \\ \frac{z}{(z-1)(z-2)} = \frac{-1}{z-1} + \frac{2}{z-1+1-2} = \frac{-1}{z-1} + \frac{2}{(z-1)-1} \\ = \frac{-1}{z-1} + \frac{2}{(z-1)[1-\frac{1}{z-1}]} = \frac{-1}{z-1} + \frac{2}{z-1} \left[1 - \frac{1}{z-1}\right]^{-1} \\ = \frac{-1}{z-1} + \frac{2}{z-1} \left[1 + \frac{1}{z-1} + \frac{1}{(z-1)^2} + \frac{1}{(z-1)^3} + \dots\right]$$

$$(b) \quad \text{Let } z = e^{i\theta} \Rightarrow dz = e^{i\theta} i d\theta \Rightarrow d\theta = \frac{dz}{iz}$$

$$\cos \theta = \frac{1}{2} \left(z + \frac{1}{z} \right)$$

$$\begin{aligned} \therefore \int_0^{2\pi} \frac{1}{2 + \cos \theta} d\theta &= \int_C \frac{1}{2 + \frac{1}{2} \left(z + \frac{1}{z} \right)} \frac{dz}{iz} = \frac{1}{i} \int_C \frac{1}{z \left(2 + \frac{z}{2} + \frac{1}{2z} \right)} dz \\ &= \frac{1}{i} \int_C \frac{1}{2z + \frac{z^2}{2} + \frac{1}{2}} dz = \frac{1}{i} \int_C \frac{2}{z^2 + 4z + 1} dz \\ &= \frac{2}{i} \int_C \frac{1}{z^2 + 4z + 1} dz \end{aligned}$$

$$\text{here } f(z) = \frac{1}{z^2 + 4z + 1}$$

$$\begin{aligned} D_x = 0 \Rightarrow z^2 + 4z + 1 = 0 \Rightarrow z &= \frac{-4 \pm \sqrt{16 - 4}}{2} = \frac{-4 \pm \sqrt{12}}{2} \\ &= \frac{-4 \pm 2\sqrt{3}}{2} = -2 \pm \sqrt{3} \end{aligned}$$

$z = -2 + \sqrt{3}$ & $-2 - \sqrt{3}$ are simple poles

$$C: |z| = 1$$

$$\begin{aligned} z = -2 + \sqrt{3}, \quad |z| &= |-2 + \sqrt{3}| = 1 - 0.268 = 0.732 < 1, \text{ lies inside } C \\ z = -2 - \sqrt{3}, \quad |z| &= |-2 - \sqrt{3}| = |-3.732| = 3.732 > 1, \text{ lies outside } C \end{aligned}$$

$$\text{By CRT, } \int_C f(z) dz = 2\pi i [\text{Sum of residues}]$$

$$\begin{aligned} \text{Res } f(z) \Big|_{z = -2 + \sqrt{3}} &= \frac{1}{2z + 4} \Big|_{z = -2 + \sqrt{3}} = \frac{1}{2(-2 + \sqrt{3}) + 4} = \frac{1}{2\sqrt{3}} \end{aligned}$$

$$\therefore \int_C \frac{1}{z^2 + 4z + 1} dz = 2\pi i \times \frac{1}{2\sqrt{3}} = \frac{\pi i}{\sqrt{3}}$$

$$\therefore \int_0^{2\pi} \frac{1}{2 + \cos \theta} d\theta = \frac{2}{i} \times \frac{\pi i}{\sqrt{3}} = \frac{2\pi}{\sqrt{3}}$$

6(a) Using Cauchy Residue Theorem, evaluate $\int \frac{z^2}{(z-1)^2(z+2)} dz$ over the circle $|z|=3$

(b) Find the Taylor series of $\frac{\sin z}{z-\pi}$ about the point $z=\pi$

(c) Evaluate $\int \frac{\sin z}{z^6}$ over the circle $|z|=2$ using Cauchy Residue Theorem.

Ans: (a) $\int_C \frac{z^2}{(z-1)^2(z+2)} dz$ where $C: |z|=3$

$Df=0 \Rightarrow (z-1)^2(z+2)=0 \Rightarrow z=1, -2$

$z=+1$ is a pole of order 2 & $z=-2$ is a simple pole.

$C: |z|=3$

$z=+1, |z|=|+1|=1 < 3$, lies inside C

$z=-2, |z|=|-2|=2 < 3$, lies inside C

By CRT, $\int_C \frac{z^2}{(z-1)^2(z+2)} dz = 2\pi i \left[\text{Res } f(z) \Big|_{z=+1} + \text{Res } f(z) \Big|_{z=-2} \right]$

$\text{Res } f(z) \Big|_{z=+1} = \frac{1}{(2-1)!} \lim_{z \rightarrow 1} \frac{d}{dz} \left[(z-1)^2 \cdot \frac{z^2}{(z-1)^2(z+2)} \right] = \lim_{z \rightarrow 1} \frac{d}{dz} \left[\frac{z^2}{(z+2)} \right]$

$= \lim_{z \rightarrow 1} \left[\frac{(z+2) \cdot 2z - z^2(1)}{(z+2)^2} \right] = \frac{(1+2)2 - 1}{(1+2)^2} = \frac{6-1}{9} = \frac{5}{9}$

$\text{Res } f(z) \Big|_{z=-2} = \frac{z^2}{(z-1)^2 + (z+2)2(z-1)} \Big|_{z=-2} = \frac{4}{9+0} = \frac{4}{9}$

$\therefore \int_C \frac{z^2}{(z-1)^2(z+2)} dz = 2\pi i \left[\frac{5}{9} + \frac{4}{9} \right] = 2\pi i \times \frac{9}{9} = \underline{\underline{2\pi i}}$

(b) $\frac{\sin z}{z-\pi} = \frac{\sin(z-\pi+\pi)}{z-\pi} = \frac{1}{z-\pi} \sin((z-\pi)+\pi)$

$\sin(A+B) = \sin A \cos B + \cos A \sin B$

$= \frac{1}{z-\pi} [\sin(z-\pi) \cos \pi + \cos(z-\pi) \sin \pi]$

$= \frac{1}{z-\pi} [-\sin(z-\pi) + 0] = -\frac{1}{z-\pi} \sin(z-\pi)$

$= \frac{-1}{z-\pi} \left[(z-\pi) - \frac{(z-\pi)^3}{3!} + \frac{(z-\pi)^5}{5!} - \dots \right] = -1 + \frac{(z-\pi)^2}{2!} - \frac{(z-\pi)^4}{4!} + \dots$

$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$

(c) Evaluate $\int_C \frac{\sin z}{z^6} dz$ where $C: |z| = 2$.

$Df = 0 \Rightarrow z^6 = 0 \Rightarrow z = 0$

$\therefore z = 0$ is a pole of order 6

Let $C: |z| = 2$

$z = 0, |z| = |0| = 0 < 2$, lies inside C

By CRT, $\int_C \frac{\sin z}{z^6} dz = 2\pi i \left[\text{Res}_{z=0} f(z) \right]$

$\text{Res}_{z=0} f(z) = \frac{1}{(6-1)!} \lim_{z \rightarrow 0} \frac{d^{6-1}}{dz^{6-1}} \left[z^6 \cdot \frac{\sin z}{z^6} \right]$

$= \frac{1}{5!} \lim_{z \rightarrow 0} \frac{d^5}{dz^5} [\sin z]$

$= \frac{1}{5!} \lim_{z \rightarrow 0} \cos z$

$= \frac{1}{5!} \times 1 = \frac{1}{5!} = \frac{1}{120}$

$\therefore \int_C \frac{\sin z}{z^6} dz = 2\pi i \times \frac{1}{120} = \frac{\pi i}{60}$

$\frac{d}{dz} [\sin z] = \cos z$

$\frac{d^2}{dz^2} [\sin z] = -\sin z$

$\frac{d^3}{dz^3} [\sin z] = -\cos z$

$\frac{d^4}{dz^4} [\sin z] = \sin z$

$\frac{d^5}{dz^5} [\sin z] = \cos z$

MODEL QUESTION PAPERS

COMPLEX ANALYSIS

1. (i) Find u & v of the following

(a) $f(z) = 5z^2 - 12z + 3 + 2i$

(b) $f(z) = \frac{1}{1+z}$

(ii) Prove that $f(z) = z\bar{z}$ is differentiable & find the derivative

Ans: (i) (a) ~~Ans~~: Refer Module - 3, pg no. 2

(b) $f(z) = \frac{1}{1+z} = \frac{1}{(1+x)+iy} \frac{(1+x)-iy}{(1+x)-iy} = \frac{(1+x)-iy}{(1+x)^2 - (iy)^2}$

$= \frac{(1+x)-iy}{(1+x)^2 + y^2} = \frac{1+x}{(1+x)^2 + y^2} + i \frac{-y}{(1+x)^2 + y^2}$

$$\therefore u = \frac{1+x}{(1+x)^2 + y^2} ; v = \frac{-y}{(1+x)^2 + y^2}$$

$$(ii) f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z+\Delta z) - f(z)}{\Delta z}$$

$$f(z) = 2z$$

$$f(z+\Delta z) = 2(z+\Delta z) = 2z + 2\Delta z$$

$$\therefore f'(z) = \lim_{\Delta z \rightarrow 0} \frac{2z + 2\Delta z - 2z}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{2\Delta z}{\Delta z} = \lim_{\Delta z \rightarrow 0} 2 = 2, \text{ exists}$$

$$\therefore f(z) = 2z \text{ is differentiable \& } f'(z) = 2z$$

2. (a) Find & sketch the images of the line $x=c$ under $w=z^2$

(b) Prove that $f(z) = \bar{z}$ is not differentiable

Ans: (a) Refer NB, Module 3, Page no. 42 & 43, Case 1

(b) Refer NB, Module 3, Pg no: 7

3(a) Integrate $z^{-1} \sin z$ around a unit circle

(b) Find the residue of $f(z) = \frac{9z+i}{z^3+z}$ at $z=i$

Ans: (a) Refer NB, Module 5, Page no. 25 [OR use Cauchy Integral formula]

$$(b) Df=0 \Rightarrow z^3+z=0 \Rightarrow z(z^2+1)=0$$

$$\Rightarrow z=0, z=i, z=-i$$

$z=i$ is a simple pole of $f(z)$

$$\therefore \text{Res } f(z) = \lim_{z \rightarrow i} (z-i) f(z) = \lim_{z \rightarrow i} (z-i) \frac{9z+i}{z(z^2+1)} = \lim_{z \rightarrow i} (z-i) \frac{9z+i}{z(z+i)(z-i)}$$

$$= \frac{9i+i}{i(i+i)} = \frac{10i}{i \times 2i} = \frac{5}{i} = \underline{\underline{-5i}}$$

4. Find $\int_c \frac{4-3z}{z^2-z} dz$ where 0 & 1 are inside c

(ii) Find the integral of $f(z) = \text{Re } z$ from 0 to $1+2i$ along the path c^* and c consisting of c_1 & c_2

Ans: $D_s = 0 \Rightarrow z^2 - z = 0 \Rightarrow z(z-1) = 0 \Rightarrow z = 0, 1$

Given it lies inside C , they are simple poles

$$\text{Res } f(z)_{z=0} = \lim_{z \rightarrow 0} z \cdot \frac{4-3z}{z(z-1)} = \frac{4}{-1} = -4$$

$$\text{Res } f(z)_{z=1} = \lim_{z \rightarrow 1} (z-1) \frac{4-3z}{z(z-1)} = \frac{4-3}{1} = 1$$

$$\begin{aligned} \text{By CRT, } \int_C \frac{4-3z}{z^2-z} dz &= 2\pi i \left[\text{Res } f(z)_{z=0} + \text{Res } f(z)_{z=1} \right] \\ &= 2\pi i [-4 + 1] = -6\pi i \end{aligned}$$

5. (i) Prove that $f(z) = z^2$ is analytic using C-R equation
(ii) Prove that $f(z) = \bar{z}$ is not analytic using C-R equation

Ans: (i) Refer NB, Module -3, pg no: 23

(ii) Refer NB, Module -3, pg no: 23

6. (i) check the conformality of $w = z + \frac{1}{z}$
(ii) check the conformality of e^z & draw the image of straight line $x=1$

Ans: (i) Refer NB, Module 3, pg no: 41

(ii) Refer NB, Module 3, pg no: 44 (Use $a=1$ in case i)

7. (i) If $f(z) = \frac{1}{z^4}$, prove that $f(z)$ is analytic or not

(ii) Find the image of left upper half with the condition ~~0 < y < \pi~~,
 $w = e^z$, $p = e^x$, $\phi = y$, $0 < y \leq \pi$

$$\begin{aligned} \text{Ans: (i)} \quad z &= re^{i\theta} \\ f(z) &= \frac{1}{(re^{i\theta})^4} = \frac{1}{r^4 e^{i4\theta}} = \frac{1}{r^4} e^{-i4\theta} \\ &= \frac{1}{r^4} [\cos 4\theta - i \sin 4\theta] = \frac{\cos 4\theta}{r^4} + i \frac{-\sin 4\theta}{r^4} \\ \therefore u &= \frac{\cos 4\theta}{r^4}, \quad v = -\frac{\sin 4\theta}{r^4} \end{aligned}$$

$$u_r = \frac{-4 \cos 4\theta}{r^5}; \quad u_\theta = \frac{-4 \sin 4\theta}{r^4}; \quad v_r = \frac{4 \sin 4\theta}{r^5}; \quad v_\theta = \frac{-4 \cos 4\theta}{r^4}$$

aka

$$u_x = \frac{1}{x} v_0 \quad \& \quad v_x = -\frac{1}{x} u_0.$$

$\therefore f(z)$ is analytic

(ii)

$$w = e^z$$

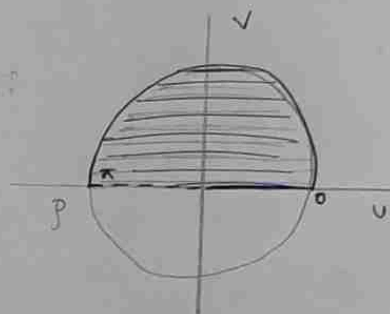
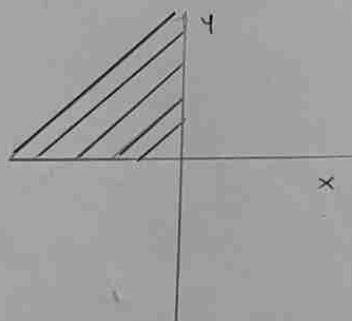
$$\text{Let } z = x+iy \quad \& \quad w = R e^{i\phi}$$

$$R e^{i\phi} = e^{x+iy} = e^x \cdot e^{iy}$$

$$\text{Comparing, } R = e^x, \quad \phi = y$$

$$\text{Given, } \rho = e^x, \quad \phi = y, \quad 0 \leq y < \pi$$

$$\therefore R = \rho, \quad 0 \leq \phi < \pi$$



8. (i) State the properties of complex line integrals

(ii) Find the integral $\int_c \sec z \, dz$ where c is the unit circle at origin.

Ans: (i) Refer NB, Module 4, pg no: 2

$$(ii) \sec z = \frac{1}{\cos z}$$

$$D_s = 0 \Rightarrow \cos z = 0 \Rightarrow z = (2n+1)\frac{\pi}{2}, \quad n \in \mathbb{Z}$$

$$\text{i.e., } z = \dots, -\frac{3\pi}{2}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}, \dots \text{ all lie outside } c.$$

$$\therefore \text{By Cauchy's theorem, } \int_c \sec z \, dz = 0$$

9. Find the ^{singularity} ~~poles~~ of $f(z) = z^{-5} \sin z$

$$\text{Ans: } f(z) = z^{-5} \sin z = \frac{\sin z}{z^5} = \frac{1}{z^5} \left[z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots \right]$$

$$\cancel{D_s = 0} \Rightarrow \cancel{z^5 = 0} \Rightarrow \cancel{z = 0} = \frac{1}{z^4} - \frac{1}{z^2 3!} + \frac{1}{5!} - \dots$$

$\therefore z = 0$ is a pole of order 4.

10. (a) Show that under the transformation $w = z^2$, the first quadrant of the z -plane is mapped to the upper half of w -plane.

(b) Show the transformation $w = e^z$

Ans: (a) First quadrant of z -plane $\Rightarrow x > 0, y > 0$

$$w = z^2$$

$$\text{Let } w = u+iv \text{ \& } z = x+iy$$

$$u+iv = (x+iy)^2 = x^2 - y^2 + i2xy$$

$$u = x^2 - y^2 ; v = 2xy$$

$x > 0, y > 0 \Rightarrow v$ takes only real values \& $v > 0$

$\Rightarrow$ upper half plane of the w -plane

(b) Refer Notebook, Module 3, pg no: 44 \& 45

11. (a) State Laplace equation \& check whether $u = x^2 - y^2 - y$ is harmonic or not.

(b) Find the image of the region $0 < y \leq \pi$ under e^z

Ans: (a) Laplace equation: $\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$ where ϕ is a function of x \& y
harmonic / not: Refer NB, Module -3, pg 37

(b)

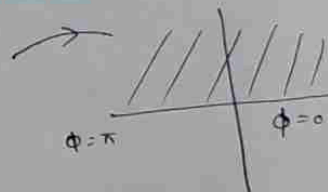
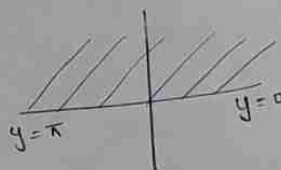
$$w = e^z$$

$$\text{Let } w = Re^{i\phi} \text{ \& } z = x+iy$$

$$\Rightarrow Re^{i\phi} = e^{x+iy} = e^x \cdot e^{iy}$$

$$\Rightarrow R = e^x, \phi = y$$

$$0 < y \leq \pi \Rightarrow 0 < \phi \leq \pi$$



12. (a) Find $\int_c \frac{z^4 - 3z^2 + 6}{(z+i)^3} dz$ where $c: |z| = 2$

(b) Find the ~~the function~~ residue of $f(z) = \frac{\sin 2z}{z^6}$

Ans: (a) $D_0 = 0 \Rightarrow (z+i)^3 = 0 \Rightarrow z+i = 0 \Rightarrow z = -i$

$$c: |z| = 2$$

$z = -i, |i| = 1 < 2$, lies inside c

$$\therefore \int_c \frac{z^4 - 3z^2 + 6}{(z+i)^3} dz = \frac{2\pi i}{3!} f'''(-i), \text{ by C.I.F}$$

$$\text{here } f(z) = z^4 - 3z^2 + 6$$

$$f'(z) = 4z^3 - 6z$$

$$f''(z) = 12z^2 - 6 \Rightarrow f''(-i) = -12 - 6 = -18$$

$$\therefore \int_c \frac{z^4 - 3z^2 + 6}{(z+i)^3} dz = \pi i \times -18 = \underline{\underline{-18\pi i}}$$

$$(b) D_0 = 0 \Rightarrow z^6 = 0 \Rightarrow z = 0$$

$z = 0$ is a pole of order 6

$$\therefore \operatorname{Res}_{z=0} f(z) = \frac{1}{(6-1)!} \lim_{z \rightarrow 0} \frac{d^{6-1}}{dz^{6-1}} [(z-0)^6 \cdot f(z)]$$

$$= \frac{1}{5!} \lim_{z \rightarrow 0} \frac{d^5}{dz^5} \left[z^6 \cdot \frac{\sin 2z}{z^4} \right]$$

$$= \frac{1}{120} \lim_{z \rightarrow 0} [32 \cos 2z]$$

$$= \frac{1}{120} \times 32 \cos 0 = \frac{32}{120}$$

$$= \underline{\underline{\frac{4}{15}}}$$

$$\frac{d}{dz} [\sin 2z] = 2 \cos 2z$$

$$\frac{d^2}{dz^2} [\sin 2z] = -4 \sin 2z$$

$$\frac{d^3}{dz^3} [\sin 2z] = -8 \cos 2z$$

$$\frac{d^4}{dz^4} [\sin 2z] = 16 \sin 2z$$

$$\frac{d^5}{dz^5} [\sin 2z] = 32 \cos 2z$$

13. (a) Evaluate $\int_0^{2\pi} \frac{1}{5 + 4 \cos \theta} d\theta$ where $c: |z|=1$

(b) Find $\int_{-\infty}^{\infty} \frac{x^2 - 1 + 2}{x^4 + 10x^2 + 9} dx$.

Ans: (a) Refer NB, Module 5, page no: 36 (Replace $\sin \theta$ by $\cos \theta$)

(b) Refer Model QP July 2017, Qno: 5(b)